UNIT - III

Handwritten Study Notes | JNTU(A) R23

Syllabus Topics:

Surface water flows: structures and channels | Groundwater - surface water exchange in wetlands | Surface water flows II and wetland hydrology case studies | Flow and mixing in wetlands | Wetland water quality: nutrients, organic/inorganic contaminants, sediments and colloids | Wetland transport models: plug flow, CSTRs and CSTRs in series | Intro to method of moments

1. Surface Water Flows: Structures and Channels

Surface water enters and leaves a wetland as channel flow (streams, ditches), overland sheet flow, tidal flow and flow through hydraulic structures. The way water is controlled decides water level, residence time and hydroperiod.

Open channel flow basics

* Free-surface flow driven by gravity; characterised by depth y, velocity V, discharge Q = A V.
* Continuity: Q = A1 V1 = A2 V2 (steady, incompressible flow).
* Uniform flow: Manning's equation for velocity.

$$V = (1/n)\, R^{(2/3)}\, S^{(1/2)} \qquad \text{(Manning)}$$

n = roughness coefficient (earth channel 0.02–0.03, vegetated wetland 0.05–0.4), R = hydraulic radius = A/P, S = bed slope. Discharge Q = A V.

* Froude number Fr = V / sqrt(g y): Fr < 1 subcritical (tranquil, typical of wetlands), Fr = 1 critical, Fr > 1 supercritical (rapid).
* Reynolds number Re = V R / nu decides laminar (< 500) or turbulent (> 2000) flow. Vegetated wetlands often have low velocities,

'laminar-transitional'.

Flow through vegetation

* Emergent plants (cattail, reed, bulrush) create drag; n is not constant - it falls as depth increases. Stem density and diameter control resistance.
* Drag force on plant: Fd = 0.5 Cd rho a V^2 (a = projected area); Cd about 1 for cylinders.
* Result: slow flow, uniform spreading (sheet flow), more settling and contact with plant surface for water treatment. But flow may short-circuit through open channels (preferential paths).

Hydraulic structures for wetland water control

Structure	Use in wetlands
Weir (sharp, broad crested, V-notch)	control and measure water level/outflow; Q = C L H^1.5 (rectangular), Q = 1.38 H^2.5 for 90 degree V-notch
Flume (Parshall)	flow measurement with small head loss, passes sediment
Culvert	pass flow under roads/embankments; inlet or outlet control
Stop-logs / gates / sluice	adjustable water level; allow drawdown and flushing
Riser pipes / standpipes	fine control of depth in constructed wetlands
Levees, berms, spillways	retain water, protect from floods, emergency overflow
Inlet and outlet manifolds	spread flow evenly across the wetland width
Fish ladders / flap gates	ecological passage, stop tidal back-flow

Formula:

Weir equation (rectangular, free flow): Q = 1.84 L H^(3/2) (SI units, m3/s).

Use contraction correction L = L0 - 0.1 n H for end contractions.

Channels in wetlands

* Natural: creeks, tidal creeks (in salt marsh), meandering sloughs; man-made: ditches, canals, cells linked by channels.
* Channel design: stable, non-eroding velocity (< 0.6 m/s earth); side slope 2:1 to 3:1; free-board; vegetation lining.
* Wetland flow paths and channels (hydroperiod, residence time) are managed by distribution of channels and berms; deep zones and open water zones are used for mixing and re-aeration.

2. Groundwater - Surface Water Exchange in Wetlands

Wetlands are interfaces between surface water and groundwater. They may be recharge, discharge or flow-through (both) types.

Type	Flow direction	Remarks
Discharge wetland	groundwater -> wetland	water table higher than wetland; springs, fens, stable water level and chemistry
Recharge wetland	wetland -> groundwater	water table below the wetland bottom; seasonal; depressions (prairie potholes)
Flow-through	in on one side, out on the other	both; typical for lakes and riparian wetlands

Darcy's law for seepage

$$Q = K A (dh/dl) \quad \text{(Darcy)}$$

K = hydraulic conductivity of the bed material (clay 1e-9 m/s; peat 1e-6; sand 1e-4), A = seepage area, dh/dl = hydraulic gradient between wetland stage and adjacent water table.

* Seepage rate per unit area: $q = K i$. Sign gives gain (+) or loss (-) in budget. Often a small term, but important in dry season and for water quality (groundwater is stable temperature and rich in minerals).
* Measurement: piezometers/mini-piezometers and nested wells (head difference), seepage meters (direct flux), temperature profiles, tracers

(Rn-222, chloride), water balance residual, numerical models (MODFLOW).

* Factors: water table slope, bed sediment (clay layer, peat), vegetation ET (daily drawdown of the water table), seasonal rainfall, pumping nearby, tides.
* Hyporheic exchange: mixing of surface water with shallow subsurface water through bed sediments - improves nutrient retention (denitrification).
* Management: do not drain or pump groundwater near wetland, maintain recharge areas in the catchment, clay liners in constructed wetlands to avoid leakage.

3. Surface Water Flows II and Wetland Hydrology Case Studies

* Sheet flow in the Everglades (Florida): very flat slope 3 cm/Km, vegetation (sawgrass) drag controls flow; compartmentalisation by canals and levees has changed the hydroperiod; Comprehensive Everglades Restoration Plan restores flow volume, timing and distribution.
* Chilika lake (Odisha, India): brackish lagoon, Ramsar site; inlet to sea was choked by silt, salinity and fish catch fell; hydraulic intervention opened new mouth (2000) -> restored salinity regime, water exchange and fisheries; weed decreased. Example of hydrology-based restoration.
* East Kolkata Wetlands (Ramsar): natural sewage-fed fishery; waste water flows through channels to ponds - low-cost treatment of about 750 million litres per day; fish and vegetables grown.
* Keoladeo Ghana National Park (Bharatpur): artificial wetland whose water is controlled by sluice gates and canals from Ajan Bund; decline in water due to diversions caused loss of bird habitat, management of inflow is vital.
* Loktak lake (Manipur): Ithai barrage on Manipur river changed hydroperiod, killed phumdis (floating biomass), submerged farms; shows effect of a structure on a wetland.

* Constructed treatment wetlands (Arcata, California; Kolkata; Coimbatore): flow, depth and residence time are designed for removing BOD, N and P.

Case study lessons:

Lessons: restore natural hydroperiod, maintain connectivity with river/sea, avoid over-regulation, monitor water level and quality, involve local community.

4. Flow and Mixing in Wetlands

Mixing decides how fast pollutant spreads and how long water stays inside the wetland (residence time), thus treatment performance.

Nominal residence time $t = V / Q$ (volume / flow rate)

* Actual residence time differs due to dead zones, short-circuiting and variations in depth: hydraulic efficiency = t_peak / t_nominal (e_v = t_mean / t_nominal).

Mixing processes

* Advection: transport by mean flow (velocity V).
* Molecular diffusion: very slow random motion, coefficient D about 1e-9 m2/s.
* Turbulent diffusion and dispersion: caused by eddies, velocity differences, wind, plants - much larger than molecular diffusion; longitudinal dispersion coefficient E.
* Vegetation increases mixing (wake of stems) but reduces velocity; open water zones have wind mixing and short-circuiting.

Advection-dispersion: $dC/dt + V\, dC/dx = E\, d2C/dx2 - KC$

C = concentration, K = first-order decay (reaction) constant. Peclet number Pe = V L / E tells which process dominates: Pe large -> plug flow like; Pe small -> well mixed.

Tracer studies

* Inject a conservative tracer (Rhodamine WT, bromide, lithium chloride, fluorescein) at the inlet and measure concentration versus time at the outlet -> RTD curve (residence time distribution), E(t).
* Mean residence time = first moment of RTD; variance = second central moment gives the spread (dispersion).
* Common patterns: early peak = short-circuit, long tail = dead zones; ideal curve = narrow peak at t = V/Q.

5. Wetland Water Quality: Contaminants

Nutrients (N and P)

* Nitrogen forms: organic N, ammonium NH_4^+, nitrite, nitrate NO_3^-, N_2 gas. Removal by plant uptake, nitrification (aerobic) then denitrification (anoxic, NO_3^- -> N_2), ammonia volatilisation, sedimentation of organic N.
* Phosphorus: dissolved ortho-P, particulate P. Removal by adsorption to soil (Fe, Al, Ca), precipitation, sedimentation, plant uptake and peat accretion. Long-term sink is limited; released again under anoxia.
* Excess N and P -> eutrophication, algal blooms, low DO.

Organic and inorganic contaminants

* Organic: BOD, COD, pesticides (DDT, atrazine), PAHs, PCBs, pharmaceuticals, oil - removed by sedimentation, microbial degradation (biofilm on plants), adsorption, photolysis and volatilisation.
* Inorganic: heavy metals (Pb, Cd, Cu, Zn, Hg, As), salts, acid mine drainage - removed by precipitation as sulphides/hydroxides, adsorption on organic matter and Fe/Mn oxides, plant uptake (phytoremediation, e.g. water hyacinth, Typha).
* Pathogens: removed by sedimentation, UV, predation, natural die-off.

Sediments and colloids

* **Sediments:** suspended solids (clay, silt) settle when velocity falls in vegetation; Stokes law for settling velocity.

$$V_s = g(\rho_s - \rho_w) d^2 / (18 \mu) \quad \text{(Stokes)}$$

* **Colloids:** particles 1 nm – 1 micron (clay, humic substances, bacteria, iron oxyhydroxides). Stay in suspension due to surface charge, carry sorbed metals, P and organics (facilitated transport), removed by coagulation/flocculation, filtration on plants and biofilm.
* Sediment burial reduces the loading but eventually deposited sediment fills the wetland and reduces storage volume and capacity.

Pollutant	Main removal in wetland
TSS	settling, filtration by plants
BOD	microbial degradation on biofilm
Nitrogen	nitrification-denitrification, uptake
Phosphorus	sorption, sedimentation, accretion
Metals	precipitation, adsorption, uptake
Pathogens	settling, UV, die-off

6. Wetland Transport Models I

Reactor models represent the wetland as an idealised reactor: the first-order decay of pollutant with rate constant K is used for design.

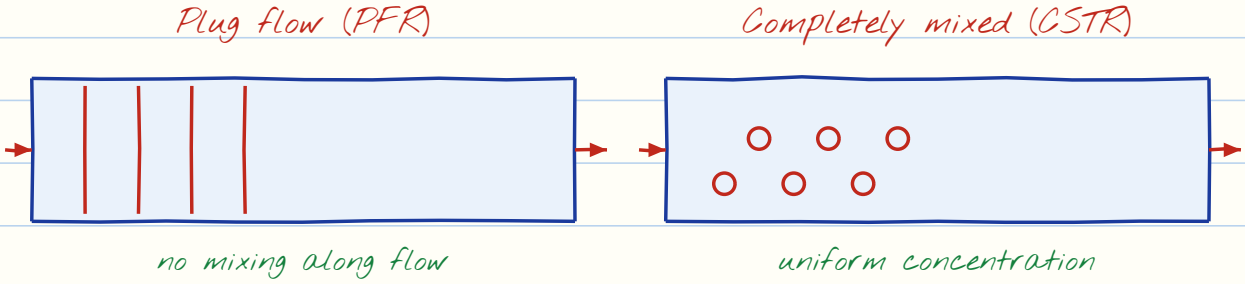


Fig 1: Two ideal reactors

A. Plug flow reactor (PFR)

* No mixing in the direction of flow; each slice of water travels as a plug; all fluid stays exactly $t = V/Q$.

$$C_{out} = C_{in} \exp(-K t)$$ (first-order)

This is the most efficient reactor: lowest outlet concentration for a given K and t. Used for long narrow wetland cells with length : width > 3:1.

B. Continuous stirred tank reactor (CSTR)

* Perfect mixing; outlet concentration = concentration inside the tank everywhere.
* Mass balance (steady): $Q C_{in} = Q C + K C V \rightarrow$

$$C_{out} / C_{in} = 1 / (1 + K t)$$

* Less efficient than PFR for the same t (because incoming clean/dirty water is instantly diluted down to the lower level where the reaction rate is low).
* Unsteady: $V dC/dt = Q (C_{in} - C) - KVC$; solution for step input $C = C_{final} (1 - \exp(-t/\tau))$ with $\tau = V/Q$.

C. CSTRs in series (tanks in series model)



$C_n / C_0 = 1 / (1 + t/N)^N$ (N tanks in series)

Fig 2: N tanks in series

$$C_N / C_0 = 1 / (1 + K t / N)^N$$

* Each tank has volume V/N; N = 1 gives a single CSTR, N -> infinity gives a PFR. So real wetlands (partial mixing) lie between: use N from tracer data.

* Relation with dispersion: variance of RTD sigma^2 = t^2 / N, so N = t^2 / sigma^2. For Pe: N = Pe / 2 (approx).
* Kadlec and Wallace tanks-in-series (TIS) model with first order area-based K: C* = (C - C*) ... P-K-C* model, includes background concentration C*.

$$(C_{out} - C^*) / (C_{in} - C^*) = (1 + K / (N q))^{(-N)} \quad \text{(P-K-C*)}$$

q = hydraulic loading rate = Q/A (m/yr), K = areal rate constant (m/yr), C* = irreducible background concentration.

> Compare:
> Efficiency order for same K and t: PFR > CSTRs in series > single CSTR. Series arrangement of cells improves performance.

7. Introduction to Method of Moments

Moments summarise a tracer response curve C(t) (or RTD) by a few numbers instead of fitting a full curve.

$$M_n = \int t^n C(t)\, dt \quad \text{(n-th absolute moment, from 0 to infinity)}$$

* Zeroth moment M0 = integral C dt = area under curve -> mass recovery (M0 Q = mass injected if conservative).
* First moment (mean) : mean residence time tm = M1 / M0 (centroid of curve).
* Second central moment (variance): sigma^2 = M2/M0 - tm^2 -> spread of the curve (dispersion).
* Third moment (skewness): asymmetry; long tail indicates dead zones or storage.

$$\text{Effective volume } V_{eff} = Q\, tm \quad ; \quad \text{dead volume fraction} = 1 - V_{eff} / V$$

* Dispersion number: sigma^2 / tm^2 = 2 (E/VL) - 2 (E/VL)^2 (1 - exp(-VL/E)) for closed vessel; N tanks: sigma^2/tm^2 = 1/N.
* Procedure: (1) inject pulse tracer, (2) sample outlet at time intervals, (3) compute C vs t, (4) numerical integration by trapezoid rule, (5)

calculate M0, M1, M2, (6) find tm, sigma^2, N, Pe.

Mini example

tm = 10 days and sigma^2 = 25 days^2 -> N = tm^2 / sigma^2 = 100 / 25 = 4 tanks. With nominal t = 12 d, effective volume ratio = 10/12 = 0.83 -> 17 % dead zone.

> Take-away:
> Method of moments is a quick way to get residence time, dispersion and number of tanks without fitting complex models.

8. Design Relations for Treatment Wetlands

* Free water surface (FWS) wetland: shallow water 0.1 - 0.6 m, emergent vegetation; area A = Q / q.
* Subsurface flow (SSF) wetland: water flows through gravel/sand bed under the surface (horizontal or vertical), less mosquito problem, smaller area but costlier media.

Item	FWS wetland	SSF wetland
Water depth	0.1 - 0.6 m open water	0.3 - 0.6 m below gravel surface
Residence time	5 - 14 days	3 - 7 days
HLR	0.01 - 0.05 m/d	0.02 - 0.1 m/d
Pros	cheap, habitat value, large volume	odour free, thermal protection, compact
Cons	mosquitoes, more area	clogging, high media cost

* Area required (first-order plug flow): A = Q ln(Cin / Cout) / (K y n) with K in per day, y = depth, n = porosity.

$$A = Q \ln(C_{in}/C_{out}) / (K_A) \quad (K_A = \text{areal rate constant, m/d})$$

* Typical aspect ratio L : W = 3:1 to 10:1 to avoid short circuiting, baffles, islands and deep zones improve mixing.

* Water depth, bed slope (0.5 – 1 %), inlet distribution, vegetation density (> 60 % cover) are the design factors, also freeboard for storm flow.

9. Solved Examples and Exam Hints

Example 1: Residence time

A wetland of area 2 ha and average depth 0.4 m with porosity 0.9 receives 800 m3/day. Find t.

$$V = 20000 \times 0.4 \times 0.9 = 7200\ m^3\ ;\ t = V/Q = 7200/800 = 9\ days$$

Example 2: Pollutant removal (PFR vs CSTR)

Cin = 40 mg/L, K = 0.3 /day, t = 9 days.

* PFR: Cout = 40 exp(−0.3 × 9) = 40 × 0.067 = 2.7 mg/L
* CSTR: Cout = 40 / (1 + 0.3 × 9) = 40 / 3.7 = 10.8 mg/L
* 3 CSTRs in series: Cout = 40 / (1 + 0.3 × 9 / 3)^3 = 40 / (1.9)^3 = 40 / 6.86 = 5.8 mg/L

So the performance order PFR (2.7) < TIS (5.8) < CSTR (10.8): a lower concentration = better.

Example 3: Manning flow

Wide vegetated channel: n = 0.2, R = 0.3 m, S = 0.0002.

$$V = (1/0.2) \times 0.3^{(0.667)} \times 0.0002^{(0.5)} = 5 \times 0.448 \times 0.01414 = 0.032\ m/s$$

Example 4: Darcy seepage

K = 1e-6 m/s, head difference 0.3 m over 3 m path, area 1000 m2: Q = 1e-6 × 1000 × 0.1 = 1e-4 m3/s = 8.6 m3/day (recharge if wetland head higher).

Likely questions

* Explain groundwater–surface water exchange in wetlands (draw discharge, recharge, flow-through types).
* Derive the CSTR and tanks-in-series equations and compare with PFR.
* Explain method of moments for a tracer test with an example.
* Discuss nutrient and contaminant removal mechanisms in wetlands.
* Case study: Chilika / Everglades / East Kolkata wetlands.

10. Quick Revision Chart

Model	Mixing	Equation (first order)	Remark
PFR	none along flow	Cout/Cin = exp(-Kt)	best efficiency
CSTR	complete	Cout/Cin = 1/(1+Kt)	lowest efficiency
TIS (N tanks)	partial	1/(1+Kt/N)^N	N = t^2 / sigma^2
Dispersion (PFD)	axial dispersion E	Peclet number Pe = VL/E	Pe = 2N approx

Water quality monitoring in wetlands

* Sampling points: inlet, intermediate, outlet, and background (reference); grab and composite samples; flow-weighted for loads.
* Parameters: field (pH, DO, temperature, EC, turbidity), lab (BOD, COD, TSS, TN, TP, NH4-N, NO3-N, coliforms, metals).
* Load = concentration x flow: L = C Q (Kg/d); removal efficiency % = (Lin - Lout) / Lin x 100; areal removal rate = (Lin - Lout) / A.
* Seasonal variation: warm season gives faster microbial removal; winter plant die-back releases nutrients; storms dilute and flush.
* Long-term issues: accumulation of sediments and metals, clogging in SSF beds, plant harvesting, mosquito and odour control.

Last line:

To remember transport models: plug = no mixing (best), CSTR = full mixing (worst), series of CSTRs = in between; method of moments gives tm, variance and N from tracer data.